

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021 - 2022 (Even Semester)

INNOVATIVE TEACHING METHOD PLAN

Degree, Semester& Branch: IVSemester B.E.ECE B
Course Code & Title: EC8652 & Wireless Communication
Name of the Faculty member: Ms.S.Harini Shriram

UNIT II			
1	10.03.2022	Introduction to Multiple access Techniques,FDMA	Think Pair share
			
2	16.03.2022	CDMA, Capacity calculations of cellular CDMA, CDMA with Multiple cells	NPTEL Video
			

3	18.03.2022	Introduction to cellular concept, frequency reuse and channel assignment	Animated Video
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4	23.03.2022	Handoff strategies, prioritizing handoffs, Practical handoff considerations	One Minute Paper
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M. Seetha devi
95361910664

Handoff strategies:-
When a mobile moves to a different cell, while conversation is in the process the MSC automatically transfers the call to a new channel belonging to the base station.

Handoff Threshold:-
minimum usable signal for acceptable voice quality.

$\Delta = P_{\text{handoff}} - P_{\text{minimum acceptable level}}$

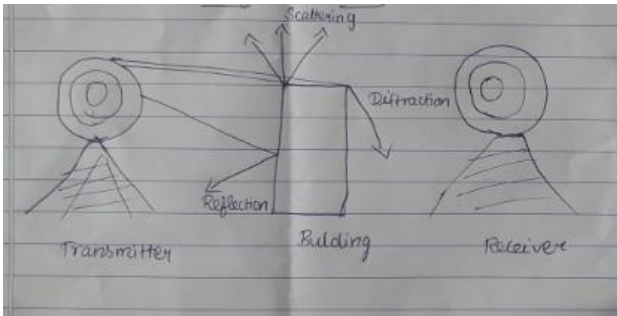
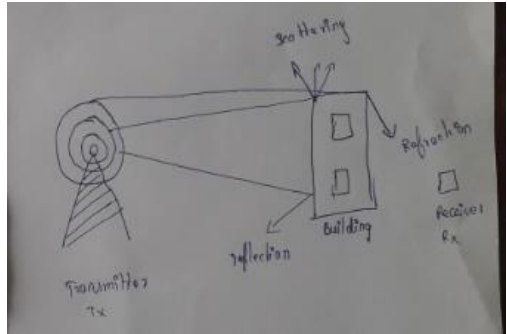
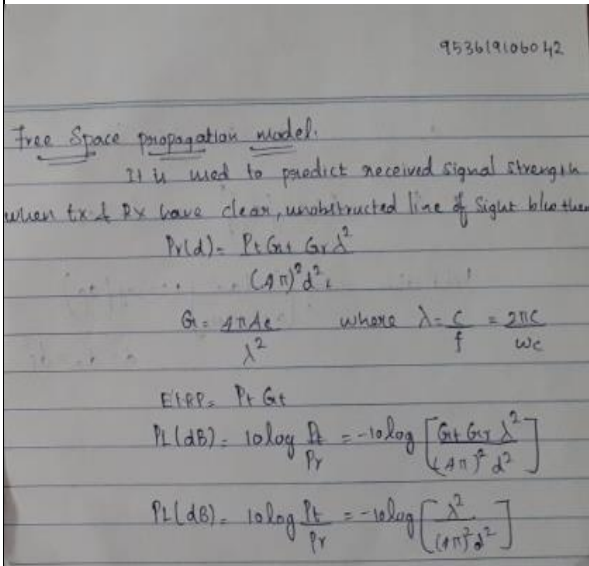
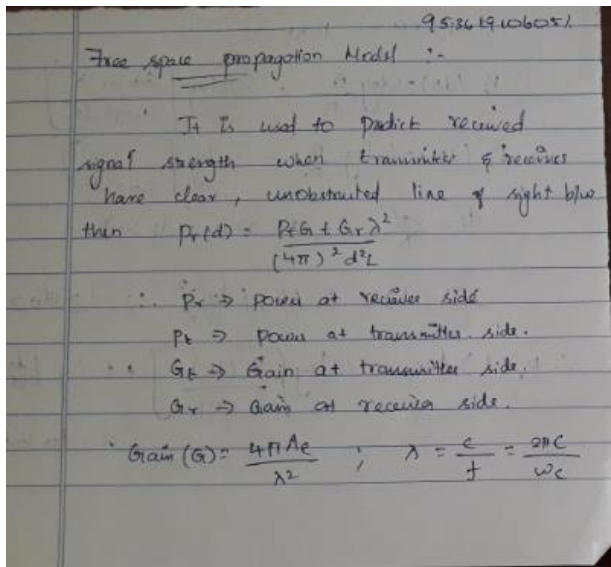
operation:-
Identifying a new base station

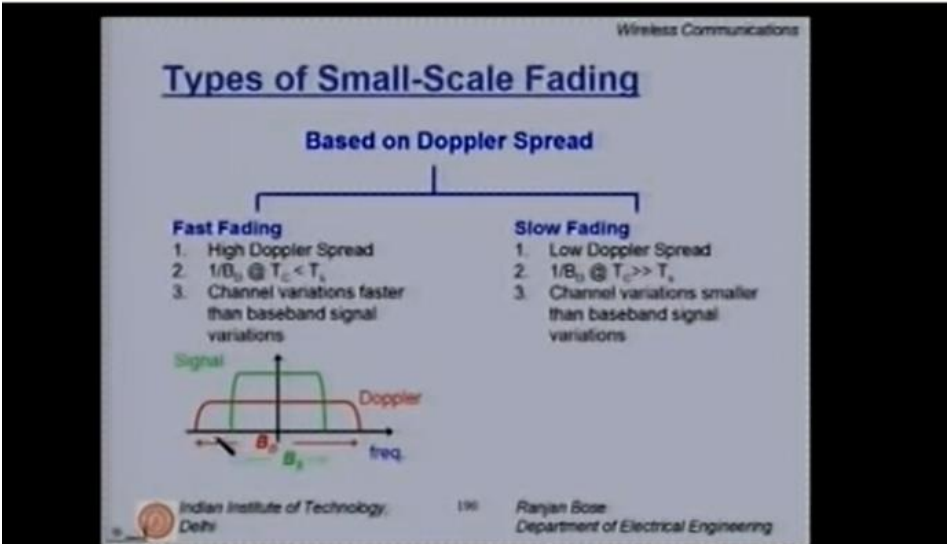
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UNIT I

1	30.3.2022	Introduction to Radio Wave Propagation, Three basic propagation mechanisms- Reflection, Diffraction, Scattering	Sketch Noting
			
2	31.3.2022	Large Scale Path Loss - Path Loss Models -Free Space Propagation Model	Concept Tests
			

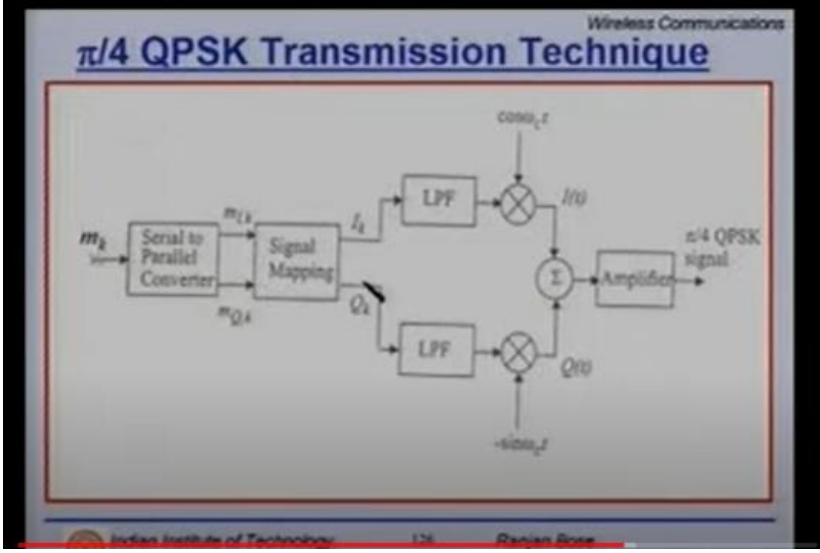
3	06.04.2022	Small scale fading and factors influencing small scale fading.	Quiz Dominos
			
4	8.4.2022	Fading due to Doppler spread-fast fading – slow fading, Revision	Passive Learning/ NPTEL Video
Lecture 19 - Frequency Flat and Frequency Selective Fading, Fast Fading, Rayleigh Fading 			

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UNIT III

1.	13.4.2022	Introduction, Structure of a wireless communication link	Quick write
		 	
2.	20.4.2022	$\pi/4$ -Differential Quadrature Phase Shift Keying	Passive Learning /NPTEL Video
			

3.	09.05.2022	OFDM Principle -Implementation of Transceivers, Frequency selective channels, Cyclic Prefix and Windowing	Flipped Class Room
			

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UNIT IV

1.	18.5.2022	Nonlinear equalization- Decision feedback equalization, MLSE equalization	Think Pair Share
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2.	20.05.2022	Diversity- Microdiversity- Spatial, Temporal, Frequency, Angle and Polarisation diversity	Passive Learning/ NPTEL Video
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Mod-01 Lec-08 Spatial Diversity and Diversity Order

diversity order of a
wired or wireline comm
system.

$$P_e(SNR) = Q(\sqrt{SNR})$$

$$\approx e^{-SNR/2}$$

40:56 / 57:20

Scroll for details

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UNIT V

1.	29.05.2022	Introduction to MIMO systems	Animated Video
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2.	3.6.2022	Transmitter diversity	Quescussion
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3.	15.6.2022	Comparison of beamforming, receiver diversity and transmitter diversity, Revision	Reciprocal Peer Questioning
			



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Innovative Practice Description

- **Unit / Topic:** III / OFDM Principle -Implementation of Transceivers, Frequency selective channels, Cyclic Prefix and Windowing
- **Course Outcome:** CO : Students will be able to analyze the performance of various digital signalling schemes of fading channels.
- **Topic Learning Outcome:** TLO: Describe the concepts of OFDM & PAPR
- **Activity Chosen:** Flipped class room
- **Justification:** It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Students are asked to prepare more in depth than before. Students are separate into groups where students are given a task to perform. Get the class back together to share the individual group's work with everyone.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PO12	PSO1	PSO3
CO3	3	2	2	3	2	2	2	2	2

• **PO / PSO mapped:**

Innovative practice	PO9	PSO1
	2	2
Justification for correlation	Quick write activity helps the students to present results as an individual, with smooth integration of contributions from all individual efforts. The course outcome is correlated at level 2.	The concepts of OFDM & PAPR techniques are used in wireless communication in order to mitigate the effects of interference and noise which occurs in communication system and thus the course outcome is mapped at level 2.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Self-learning and accountability of the students got improved. They learn how to communicate with team members and work together to achieve common goals.

- ❖ ***Benefit of the practice:***

Every student got equal opportunity to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is that few students not exposed to flipped class room.

Students struggle with self-discipline and may turn up to class without having absorbed the lesson. Few students hesitated to come forward and to interact. I have encouraged them to come forward and share their points

References:

- ❖ https://www.rcboe.org/cms/lib/GA01903614/Centricity/Domain/15451/Flip_your_Classroom.pdf
- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.
- ❖ Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from <https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html>.